Basic Properties and Substitutions

Standard forms and algebraic rearrangement:

- Polynomial long division might be required to achieve a standard form.
- Algebraic re-arrangement (completing the square) may be required before or after substitution.

1. Calculate the following using standard forms or linear substitution:

   a. $\displaystyle\int \frac{2x+1}{2-x}dx$

   b. $\displaystyle\int (2x+1)\sqrt{x-1}dx$

   c. $\displaystyle\int \frac{1}{x^2+2x+2}dx$

The Reverse Chain Rule

Rule: $\displaystyle\int f[g(x)]g'(x)dx = F(g(x)) + C$. **Special case:** $\displaystyle\int \frac{g'(x)}{g(x)}dx = \log_e|g(x)| + C$.

1. Find the anti-derivative of:

   a. $\displaystyle\int \frac{e^x}{1+5e^{2x}}dx$

   b. $\displaystyle\int \frac{\sin(x)}{\sqrt{1+\cos(x)}}dx$

2. **(Unleash the Beast):** Find $\displaystyle\int \sqrt{1-\cos(x)}dx$ for $0 \leq x \leq \pi$.

Trigonometric Identities

Case 2: Odd powers. Substitute $\sin^2(kx) = 1 - \cos^2(kx)$ or $\cos^2(kx) = 1 - \sin^2(kx)$.
Case 3: Even powers. Use double angle formulae:

$$\cos^2(A) = \frac{1}{2}(\cos(2A)+1) \text{ and } \sin^2(A) = \frac{1}{2}(1-\cos(2A))$$

1. Calculate $\int \cos(x)\sin(x)dx$ in three different ways and compare.

2. Evaluate $\int \sin^2(x)\cos^4(x)dx$.

Rational Functions and Partial Fractions

Partial Fraction Forms:

- Non-Repeated Linear Factor: $\dfrac{A}{x-\alpha}$
- Repeated Linear Factor: $\dfrac{A_1}{x-a} + \dfrac{A_2}{(x-a)^2} + \dots$
- Irreducible Quadratic: $\dfrac{Bx+C}{ax^2+bx+c}$

1. Compare finding $\displaystyle\int \frac{5x-1}{(2x+1)^2}dx$ using partial fractions versus using substitution.

2. Find the anti-derivative of $\displaystyle\int \frac{x^2}{9-x^2}dx$.

Integration by Parts

Formula: $\displaystyle\int u\,dv = uv - \int v\,du.$
Priority (u): Polynomial (Case 1) or Inverse Transcendental (Case 2).

1. Find $\displaystyle\int x\sin(x)dx$.

2. Find the single inverse transcendental integral $\displaystyle\int \arcsin(x)dx$.

Definite Integrals and Reduction Formulae

- Always substitute for the integral terminals when using a change of variable.

1. Let $I_n = \int \sin^n(x)dx$. Use integration by parts to obtain the reduction formula:

$$I_n = -\frac{1}{n}\cos(x)\sin^{n-1}(x) + \left(\frac{n-1}{n}\right) I_{n-2}$$

2. Find exact values of k and p such that $\displaystyle\int_k^4 \frac{1}{x^3 - 2x^2 + x}dx = \frac{8}{3} - \log_e(p)$.

Graphing Relationships

- If $f(a) = 0$ (x-intercept), then $F(x)$ has a stationary point at $x = a$.
- If $f(x)$ has a turning point at $x = a$, then $F(x)$ has a point of inflection at $x = a$.